# Failure (make it confidently wrong, on purpose) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint builds toward the full solution.


- Hint 1 — Finding a good question, and phrasing it to invite confidence
- Hint 2 — Running it, and explaining why it happened


## Hint 1 — Finding a good question, and phrasing it to invite confidence

Three reliable categories to pick from:

1. **A real but obscure fact** — a statistic, a minor historical detail, a small company's founding date — something genuinely true but unlikely to appear often (or precisely) in training data.
2. **A plausible-sounding but fictional entity** — ask about a "paper," "law," or "person" you invent, phrased as if it definitely exists. The model often plays along rather than questioning the premise.
3. **A request for a specific citation** — "give me the exact page number where X said Y" — models are notoriously unreliable at exact citations because that's an extremely precise fact to have memorized correctly.

The framing matters as much as the category: "What was the exact attendance at [small, real but obscure event]?" works better than "do you happen to know roughly how many people attended...?" — the second version invites hedging, the first invites confidence, because a confidently-phrased question pulls the model toward a confidently-phrased continuation.

Pick one category and write the exact prompt you'll use before moving to Hint 2.

Notice all 3 categories share one property: the question has the *grammatical shape* of something with one exact, checkable answer. "What was the attendance at X" has the same shape whether X is a famous event with a well-documented number or a made-up local parade — the model responds to the shape of the question at least as much as to whether it actually knows the answer. Before running your prompt, predict: will the model hedge ("I'm not certain, but..."), refuse ("I don't have information about..."), or answer confidently and specifically? Write your prediction down —

comparing it to what actually happens is the real test of whether you understand the mechanism, not just the trick.

## Hint 2 — Running it, and explaining why it happened

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prompt category chosen: fictional citation
example: "In which paragraph of [a real, well-known book] does the
          author first mention [a specific, invented detail]?"

expected result: the model names a specific paragraph or chapter,
                 stated confidently, that is very likely fabricated
                 — because it's answering "what would a citation for
                 this look like," not verifying the citation is real
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Run your own prompt, record the exact answer you got, and note how confidently it was phrased — that confidence is the actual finding here, not just whether the fact was wrong.

Frame your explanation paragraph around the actual mechanism: the model generates the token sequence that's statistically most likely to follow your prompt, and a confident, specific-sounding, well-formed answer is a very natural-looking continuation of a confidently-phrased factual question — regardless of whether the underlying claim is grounded in anything real.

Write your explanation now. There is no separate step where the model checks its own answer against the world — fluency and correctness are produced by the identical process and look identical from the outside, which is why this "can't just be fixed like a bug" the way one faulty line of logic can. Grounding the answer in retrieved, real source text (RAG, covered properly in Doc08) is the main real mitigation — worth naming in your explanation if you want to go one step further than the plain mechanism.

Compare your explanation against the Solution.

Full solution: Show me the solution